TECHNICAL REPORT: NETWORK DESIGN AND IMPLEMENTATION (PROJECT CLI-127)

Course: CMPG 325 Computer Networks

Client ID: CLI-127

Author: Kamogelo Semara

Student Number: 36233137

Institution: North-West University (Mafikeng Campus)

Date: 16 October 2026

1. Executive Summary

This technical report details the design, configuration, and security implementation of an enterprise-grade network infrastructure for Client CLI-127. Built around a Hierarchical Extended Star Topology, the network integrates a Cisco 3650 Multilayer Core Switch (`Core-SW`) and four Cisco 2960 Access Switches (`Studio-SW`, `Admin-SW`, `Server-SW`, and `Floor2-SW`). Address allocation is based on a Variable Length Subnet Masking (VLSM) scheme derived from the `172.30.82.0/23` parent block. Link aggregation is provided by IEEE 802.3ad LACP EtherChannels across dual Gigabit uplinks, while Layer 3 Inter-VLAN routing and Extended Access Control Lists (ACL) enforce strict server room access security. All configurations and test scenarios were successfully validated in Cisco Packet Tracer.

2. Network Topology and Architecture

The network follows the industry-standard Cisco three-tier hierarchical model (collapsed core/distribution architecture) to ensure scalability, fault isolation, and bandwidth aggregation.

2.1 Hardware and Link Allocation

- **Core Switch (`Core-SW`):** Cisco WS-C3650-24PS serving as the Layer 3 routing backbone and central ACL enforcement point.
- **Access Switches:** Four Cisco Catalyst 2960 switches handling Layer 2 access for department workstations:
    - `Studio-SW` (VLAN 10)
    - `Admin-SW` (VLAN 20)

    - `Server-SW` (VLAN 30)
    - `Floor2-SW` (VLAN 40)
- **Trunk Aggregation:** Dual physical Gigabit Ethernet links connect each access switch to `Core-SW`.
- **End Devices:** Six workstations (`Studio-PC1`, `Studio-PC2`, `Admin-PC1`, `Admin-PC2`, `Floor2-PC1`, `Floor2-PC2`) and one high-availability server (`Main-Server`).

3. VLSM IP Addressing Scheme

The addressing architecture is derived from the assigned `/23` prefix (`172.30.82.0/23`, covering `172.30.82.0` to `172.30.83.255`), offering usable IP addresses tailored to specific host density requirements.

3.1 Subnet Breakdown

1. **Studio Department (VLAN 10):**
    a. Subnet: `172.30.82.0/25`
    b. Subnet Mask: `255.255.255.128`
    c. Usable Range: `172.30.82.1` – `172.30.82.126`
    d. Default Gateway (SVI): `172.30.82.1`
    e. Host Example: `Studio-PC1` (`172.30.82.10`, Mask: `255.255.255.128`)
2. **Administration Department (VLAN 20):**
    a. Subnet: `172.30.82.128/25`
    b. Subnet Mask: `255.255.255.128`
    c. Usable Range: `172.30.82.129` – `172.30.82.254`
    d. Default Gateway (SVI): `172.30.82.129`

e. Host Example: `Admin-PC1` (`172.30.82.140`, Mask: `255.255.255.128`)

3. **Server Room (VLAN 30):**
    a. Subnet: `172.30.83.0/26`
    b. Subnet Mask: `255.255.255.192`
    c. Usable Range: `172.30.83.1` – `172.30.83.62`
    d. Default Gateway (SVI): `172.30.83.1`
    e. Host Example: `Main-Server` (`172.30.83.10`, Mask: `255.255.255.192`)
4. **Floor 2 Expansion (VLAN 40):**
    a. Subnet: `172.30.83.64/26`
    b. Subnet Mask: `255.255.255.192`
    c. Usable Range: `172.30.83.65` – `172.30.83.126`
    d. Default Gateway (SVI): `172.30.83.65`
    e. Host Example: `Floor2-PC1` (`172.30.83.70`, Mask: `255.255.255.192`)

4. Technical Implementation & Security

4.1 LACP Link Aggregation

To achieve link redundancy and aggregate trunk capacity, IEEE 802.3ad Link Aggregation Control Protocol (LACP) in active mode was configured across all uplinks:

- `Port-Channel 1 (Po1)`: Links `Studio-SW` (`Gig1/0/1`, `Gig1/0/2`) to `Core-SW`.
- `Port-Channel 2 (Po2)`: Links `Admin-SW` (`Gig1/0/3`, `Gig1/0/4`) to `Core-SW`.
- `Port-Channel 3 (Po3)`: Links `Server-SW` (`Gig1/0/5`, `Gig1/0/6`) to `Core-SW`.
- `Port-Channel 4 (Po4)`: Links `Floor2-SW` (`Gig1/0/7`, `Gig1/0/8`) to `Core-SW`.

4.2 Extended Access Control List (ACL)

Per client requirements, only the Administration subnet (`172.30.82.128/25`) is permitted to communicate with the `Main-Server` (`172.30.83.10`). All other subnets are explicitly denied server access while maintaining general inter-VLAN routing for non-server traffic.

ACL Configuration on Core-SW:

```
ip access-list extended RESTRICT_SERVER_ACCESS
 permit ip 172.30.82.128 0.0.0.127 172.30.83.0 0.0.0.63
 deny ip any 172.30.83.0 0.0.0.63
 permit ip any any
exit

interface vlan 30
 ip access-group RESTRICT_SERVER_ACCESS out
```

Placement Justification: Applying the Extended ACL outbound on Interface VLAN 30 ensures that filtering occurs at the final Layer 3 hop before entering the Server Room subnet, allowing efficient inter-VLAN routing while blocking unauthorized packets at the edge.

5. Verification & Testing Results

Source Host	Destination Host	Command Executed	Expected Result	Actual Result	Status	
EtherChannel Verification	Core-SW	Access Switches	show etherchannel summary	All Port-Channels display SU and ports P	Po1 to Po4 active and bundled	**PASS**
Inter-VLAN Communication	Admin-PC1 (172.30.82.140)	Studio-PC1 (172.30.82.10)	ping 172.30.82.10	Successful reply, 0% packet loss	Reply received, 0% loss	**PASS**
Authorized Server Access	Admin-PC1 (172.30.82.140)	Main-Server (172.30.83.10)	ping 172.30.83.10	Successful reply, 0% packet loss	Reply received, 0% loss	**PASS**
Blocked Server Access	Studio-PC1 (172.30.82.10)	Main-Server (172.30.83.10)	ping 172.30.83.10	Traffic blocked, Destination host unreachable	Destination host unreachable	**PASS**

6. Conclusion

The network design for Client CLI-127 fully satisfies all technical requirements. The VLSM scheme eliminates address redundancy, LACP EtherChannels provide robust link failover and aggregated trunk throughput, and the Extended Access Control List strictly enforces server access control. The solution is validated and completely documented for deployment.

https://github.com/kamogelosemara-a11y/Computer-Science-Project-Brief.git